Teaching with the Evolutionary Laboratories

Routes, timings and objectives for all thirteen rooms

Every room has a printable guide of its own, linked from the room's *Start here* panel. Each guide opens with learning objectives and suggested routes, then gives a **core path** sized for a BSc practical and an **extension** for MSc students. Every step asks for a prediction before the run and an explanation after it, and every guide ends with discussion questions and a page of notes for checking the reasoning. The table below is the whole set at a glance; times are minutes of active work, not counting a class discussion.

ROOM	WHAT IT IS FOR	CORE BSC	EXT. MSC
PART I — WHAT A MUTATION IS			
01 **The Copying Room** guides/01-copying-room.pdf	Copying is imperfect and undirected: lineages change, and change in different directions, with no selection at all.	30	+28
02 **The Branching Room** guides/02-branching-room.pdf	Lineages split and diverge; a tree rebuilt from final appearances alone often gets the history wrong.	28	+28
PART II — WHAT BECOMES OF ONE MUTATION			
03 **The Fate Room** guides/03-fate-room.pdf	A new neutral mutation starts as one copy in N and is usually lost within a few generations; it survives with probability 1/N.	31	+30
04 **The Drift Room** guides/04-drift-room.pdf	Chance alone moves an allele frequency, fastest in small populations; drift is unpredictable at one locus and predictable on average.	35	+34
05 **The Effective Size Room** guides/05-effective-size-room.pdf	Real populations drift like smaller ideal ones: the harmonic mean of a changing size, the rarer sex and inbreeding all set N_e below the census.	35	+30
PART III — NEUTRAL OR SELECTED			
06 **The Selection Room** guides/06-selection-room.pdf	Selection tilts the odds, drift still draws the sample; what matters is N·s, and a rare recessive advantage is nearly invisible.	35	+32
07 **The Mutation Room** guides/07-mutation-room.pdf	Selection decides a mutation's fate but never its origin: no value of s resizes a slice of the mutation wheel.	33	+32
08 **The Migration Room** guides/08-migration-room.pdf	Drift pulls two populations apart, gene flow pulls them together; what decides it is the number of migrants Nm, not the fraction m.	35	+30
09 **The Adaptation Room** guides/09-adaptation-room.pdf	Selection adds directed change on top of drift; unrelated lineages in one habitat converge, and population size decides how well mutations are sorted.	35	+30
10 **The Speciation Room** guides/10-speciation-room.pdf	Two isolated populations accumulate fixed differences, mostly by drift at loci nobody selects; selection sorts its loci faster but they are few.	37	+32
PART IV — THE MACHINERY UNDERNEATH			
11 **The Reproduction Room** guides/11-reproduction-room.pdf	Asexual reproduction doubles numbers but copies genotypes; sex pays a two-fold cost and makes genotypes that never existed before.	25	+28
12 **The Hardy–Weinberg Room** guides/12-hardy-weinberg-room.pdf	Under random mating one allele frequency fixes all three genotype frequencies in a single generation; drawing real gametes adds drift.	32	+28

ROOM	WHAT IT IS FOR	CORE BSC	EXT. MSC
13 **The Linkage Room** guides/13-linkage-room.pdf	An association between linked loci decays only as fast as recombination breaks it; while it lasts, two good mutations interfere.	35	+30
All rooms, minutes		**426**	**+392**

HOW THE GUIDES ARE MEANT TO BE USED

Students write a prediction before they press anything, then explain whether the run matched. A wrong prediction they can account for is worth more than a right one they guessed, and is the most useful thing to mark.

- **In class:** print the core-path pages; hold back the last page (*Checking your reasoning*) until the discussion, or print without it.
- **Independently:** the whole guide works as self-study, with the last page as the check.
- **For assessment:** the prediction–explanation pairs and the worksheet questions (tagged BSc or MSc) make short written tasks; the extension steps make good MSc problem-sheet questions.

A BSC COURSE IN SIX PRACTICALS OF ABOUT TWO HOURS

- **Practical 1 — mutation and divergence:** Copying (01) and Branching (02).
- **Practical 2 — the fate of a mutation:** Fate (03) and Drift (04), with Effective Size (05) steps 1–2 if time allows.
- **Practical 3 — selection and its limits:** Selection (06) and Mutation (07).
- **Practical 4 — populations in contact:** Migration (08) and Hardy–Weinberg (12).
- **Practical 5 — adaptation:** Adaptation (09), with Linkage (13) part 1 if time allows.
- **Practical 6 — speciation:** Speciation (10), with Reproduction (11) as the opening demonstration.

AN MSC MODULE

- Set each room's core path as preparation before class, with the *Checking your reasoning* page.
- Spend contact time on the extension steps, which ask for calculations to be made before the run and checked against it.
- The MSc questions of Selection, Migration, Speciation and Linkage make a good written assignment on the interplay of drift, selection and gene flow.

SINGLE SESSIONS

- **Drift and selection in 90 minutes:** Drift (04) and Selection (06) core paths.
- **Evolution is not progress (one hour, first-year):** Copying (01) steps 1–4 and Mutation (07) steps 1–4.
- **Population genetics revision (two hours):** Hardy–Weinberg (12), Drift (04) and Migration (08) core paths.
- **Why sex? (one hour):** Reproduction (11) and Linkage (13) steps 3 and 7.
- **Conservation genetics (90 minutes):** Drift (04) step 2, then the Effective Size (05) core path, then Migration (08) core path.